

Mitsue AI Data Center - Phase-1 CHP Pilot Sizing (50 kWe, Year 1-3)

How much forest to clear-cut for a 50 kWe biomass CHP prototype, with the current crew plus one hire

Working draft · v1.0 · 2026-09-12

Early planning-stage estimate. Grade-split ratio is a non-Mitsue proxy study; crew productivity is a generic benchmark. Confirm with a stand cruise and the co-op before sizing capex.

Phase-1 target: ~12.5 ha/yr clear-cut, ~37 ha over 3 years (conservative case: only C+D grade wood, ~24% of harvest, is CHP fuel). Current 2-person crew plus 1 hire covers only ~28% of the volume this needs - the gap is the point of Phase-1.

- Fuel path

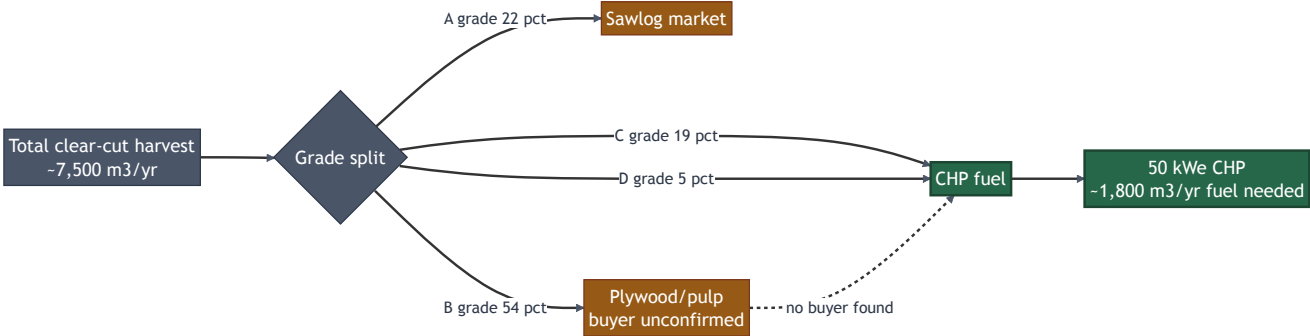
Sold elsewhere

Workforce
- Mechanized

Dashed = open question

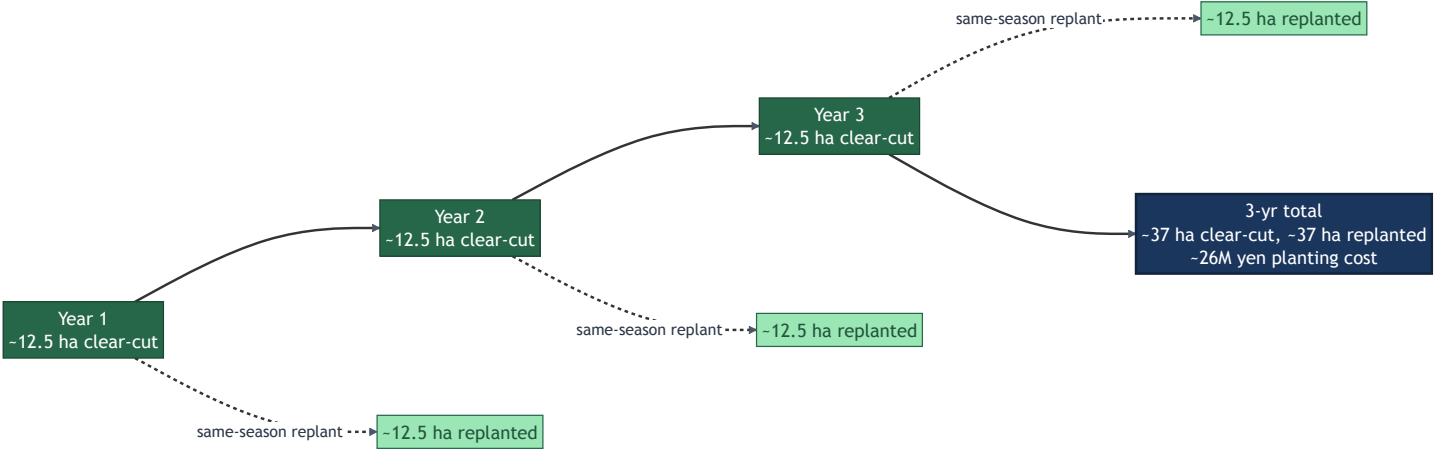
1. Fuel Flow - A/B/C/D Grade Split

Only chip/fuel-grade wood (C+D) is a confirmed CHP feedstock; B-grade buyer status is unconfirmed



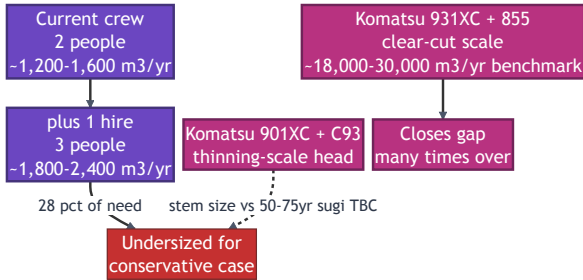
2. Three-Year Clear-Cut and Replant Ramp

Replanting phased same season as felling, matching pace - no lag



3. Workforce vs Mechanized Options

Generic productivity benchmarks, not Komatsu- or Mitsue-confirmed



4. Open Questions Driving the Scenario

Two unresolved answers set which Phase-1 case actually applies

